

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative modeler with experience developing and validating statistical and machine-learning models for financial risk, insurance pricing, and large dependent datasets. Combines econometrics, stochastic processes, likelihood-based inference, simulation, and model diagnostics with reproducible Python/PyTorch workflows on AWS and HPC. Translates business and monitoring questions into transparent model specifications, performance evidence, and actionable recommendations.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Platforms: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), R, SQL, C++/pybind11 integration, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing, cached artifacts, and restartable analytical pipelines

Statistical and Econometric Modeling: Linear and logistic regression, GLM/GBM, time-series and spatiotemporal modeling, panel/longitudinal and cross-sectional analysis, fixed effects, propensity-score matching, machine learning, MLE/GLS, Monte Carlo simulation, hypothesis testing, model selection, validation, sensitivity analysis, and performance diagnostics

QUANTITATIVE INDUSTRY EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed and validated a Sequential Probability Ratio Test for binary risk indicators, translating a monitoring problem into explicit hypotheses, likelihood-ratio boundaries, and controlled sequential decisions.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and simulation to quantify Type I/II error, boundary-crossing probabilities, stopping behavior, uncertainty, and early-decision tradeoffs; documented assumptions, performance, and limitations.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ property-risk records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Evaluated predictive performance and feature-attribution patterns and presented pricing and risk-segmentation recommendations to business stakeholders.

RESEARCH AND QUANTITATIVE MODELING

Longitudinal Observational Study - EITC and Household Labor Supply

2021

- Analyzed the relationship between Earned Income Tax Credit exposure and household labor supply using panel data, fixed-effects regression, and propensity-score matching with household income and demographic covariates; evaluated comparison-group construction, confounding assumptions, and interpretation of results.

Ph.D. Dissertation Research - Scalable Model Development and Diagnostics

Sep 2024 - Present

- **Large-Scale Model Development:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with ordered conditional models to fit up to 145,008 observations per day.
- **Data Engineering:** Built quality filters, missingness tracking, time-specific coordinate offsets, nearest-neighbor center matching, and spatial-threshold regular-grid construction for incomplete, irregular data.
- **Model Diagnostics and Monitoring:** Designed scale- and frequency-resolved tools to distinguish systematic data artifacts and measurement noise from model misspecification and identify where assumptions fail.
- **Research Engineering:** Implemented reusable Python/PyTorch workflows with numerical optimization, CPU/GPU execution, cached artifacts, and restartable HPC jobs for estimation, simulation, and diagnostics; integrated compiled C++ ordering routines through pybind11.

ADDITIONAL

Coursework: Machine Learning, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)